Lecture Notes on

General Relativity

Owen Lin

owen2006lin@gmail.com

These notes cover general relativity at the undergraduate level. The sources used are:

- Schutz, *A First Course in General Relativity*
- Carroll, *Spacetime and Geometry*

Potential sources to consult in the future:

- Hartle, *Gravity: An Introduction to Einstein's General Relativity*
- Misner, Thorne, Wheeler, *Gravitation*
- Weinberg, *Gravitation and Cosmology*
- Wald, *General Relativity*

Contents

1 Preliminaries

1.1 Coordinate Transformations

First, we'll briefly recap special relativity using index notation.

- Let's define a four vector x^μ where x^0 is the temporal component and x^i ($i \neq 0$) are the spacial coordinates. In these notes, we'll write the representation of a vector in coordinate transformation as $x^{\mu'}$ (each source has their own convention).

- The matrix to transform the vector x^ν into the new primed coordinates is

$$x^{\mu'} = \Lambda^{\mu'}{}_{\nu} x^\nu$$

  An object that transforms like vector components is said to transform **contravariantly**. Note we are using the Einstein convention where summations over repeated indices drop the Σ.

- A Lorentz Transformation is one that keeps the spacetime interval constant and fix the origin so this requires $\eta = \Lambda^T \eta \Lambda$. In index notation, this is:

$$\eta_{\alpha\beta} = \Lambda^{\mu'}{}_{\alpha} \Lambda^{\nu'}{}_{\beta} \eta_{\mu'\nu'}$$

  Note that when writing the equation using matrices, the order is important because of the limitations of matrix multiplication. However, the order is irrelevant in index notation. These matrices are **Lorentz Transformations** and are elements of the **Lorentz Group**.

- A brief aside, these matrices may seem very similar to the rotation group SO(3). The Orthogonal Group is all 3x3 matrices that satisfy $\mathbf{R^T R} = \mathbf{1}$ (SO(3) then are all special matrices in O(3), ie determinant 1). The difference between O(3) and the Lorentz group is that instead of the identity matrix, it is equal to the metric η. The identity matrix is simply the metric in **Euclidean Space**, and η is similarly the metric for a **Lorentzian Space**.

- Lorentz transformations come in two forms: rotations and boosts. Below is an example of a boost in the x direction:

$$\Lambda^{\mu'}{}_{\nu'} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  The transformed quantity for t' and x' are then $t' = t\cosh\phi - x\sinh\phi$ and $x' = -t\sinh\phi + x\cosh\phi$. The velocity follows as $v = \tanh\phi$, and we can then rewrite in more familiar form

$$t' = \gamma(t - vx)$$

$$x' = \gamma(x - vt)$$

  where $\gamma = 1/\sqrt{1-v^2}$. From here, we can proceed and obtain time dilation, length contraction, etc.

- Another transformation that preserves the spacetime interval is a pure translation: $x^\mu \to x^{\mu'} = (x^\mu + a^\mu)$ The set of all translations form an Abelian group but generally Lorentz transforms do not commute so the Lorentz Group is nonabelian. The set of all Lorentz transforms and translations form a larger nonabelian group: the **Poincaré Group**.

Now we'll take a look at how vectors and their components transform under a change of coordinates.

- Suppose in an arbitrary vector space we have a basis of vectors $\hat{e}_\mu$. Then any arbitrary vector can be written as $A = A^\mu \hat{e}_\mu$ where A^μ are the components of the vector. Often we will just call A^μ the vector for shorthand.

- An example is the tangent vector $V^\mu = \frac{dx^\mu}{d\lambda}$. Under a lorentz transformation the components change by $\frac{dx^{\mu'}}{dx^\mu}$ by the chain rule so the components transform as $V^\mu \to V^{\mu'} = \Lambda^{\mu'}{}_\nu V^\nu$ but the vector itself is invariant under transformation. So we must have $V^\mu \hat{e}_\mu = V^{\nu'} \hat{e}_{\nu'} = \Lambda^{\nu'}{}_\mu V^\mu \hat{e}_{\nu'}$. But this must hold for any vector no matter the components so we must have $\hat{e}_\mu = \Lambda^{\nu'}{}_\mu \hat{e}_{\nu'}$. This means basis vectors transform opposite to their components.

- Note that applying a transformation and then its inverse should result in the identity. In index notation, this means $\Lambda^\mu{}_{\nu'} \Lambda^{\nu'}{}_\rho = \delta^\mu_\rho$

- We can define a vector space dual to our previous vector space, which is the set of all linear maps from the original vector space to $\mathbb{R}$. We can write a set of basis dual vectors as $\hat{\theta}^\nu(\hat{e}_\mu) = \delta^\nu_\mu$. Then every dual vector can be written in component form $\omega = \omega_\mu \hat{\theta}^\mu$. (we also refer to dual vectors as "1-forms").

- The components of dual vectors transform like basis vectors (that is, inverse to vector components)
$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'} \omega_\nu$$
An object that transforms like this is said to transform **covariantly**. Meanwhile, dual basis vectors transform like vector components $\hat{\theta}^{\rho'} = \Lambda^{\rho'}{}_\sigma \hat{\theta}^\sigma$. The key takeaway is that the location of the index tells us how the object transforms. Upper index transform like vector components, while lower transforms inverse to that. Accordingly, scalars are invariant under transformation.

- An example is the gradient, with components $\frac{\partial}{\partial x^\mu}$. Under transformation, this becomes $\frac{\partial}{\partial x^{\mu'}} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial}{\partial x^\mu}$ by the chain rule, but this is equivalent to just $\Lambda^\mu{}_{\mu'} \frac{\partial}{\partial x^\mu}$. Since the gradient is a dual vector, we tend to write it with lower index as ∂_μ.

We'll later generalize this idea of transformation under change of coordinates to any transformation, not just Lorentz transformations.

1.2 Equivalence Principles

The equivalence principles are important in that they describe gravity is a property of spacetime's geometry, not the result of any additional field.

- The **Weak Equivalence Principle** states that inertial mass and gravitational mass of any object are equal.
$$m_i = m_a$$

This also means that objects traveling through some gravitational potential Φ have the same acceleration $\mathbf{a} = -\nabla\Phi$. A path through spacetime where particles don't feel any external force is called inertial, and all particles will locally travel this same path, a fact that isn't true for other forces (eg; such paths don't exist for EM).

- The **Einstein Equivalence Principle** states in small regions of spacetime, the laws of physics reduce to that of special relativity and its impossible to detect the existence of gravitational field with local experiments. For example, a person standing on the Earth and Jupiter's surface will see differences in their experiments. But if instead this person were in a sealed container travelling uniformly through the gravitational field, there would be no way to discern the two apart. Note this is only true if the box covers a small, local region of spacetime. In larger regions, tidal forces come into play which can be detected.
- It is still possible to have a theory that violates the Einstein principle but satisfies the weak equivalence principle. For example, a theory of gravity that all particles start rotating as they travel the same path in a gravitational field satisfies the weak equivalence principle but violates the Einstein principle, since you could detect a gravitational field this way.
- The **Strong Equivalence Principle** extends the Einstein Equivalence principle to massive particles, and particles that exert gravitational force on themselves, such as planets and black holes.
- An important corollary of the equivalence principles is that the acceleration due to gravity is not well defined. There's no such thing as a neutral object that we could measure the acceleration with respect to. It makes more sense then, to define freely falling through a gravitational field as being unaccelerated.
- Locality is very important to keep in mind. In special relativity we define a inertial frame by assigning every point in spacetime a clock and then connecting them with rigid rods, however this is impossible in the presence of gravity, because we defined freely falling objects as not accelerating, but they clearly are accelerating with respet to such a reference frame. Hence, we can only define such an inertial frame locally.

Example. The most famous example demonstrating the equivalence principle gravitational redshift. Consider two boxes each a distance z apart in free space and nonrelativistically moving with constant acceleration a in the same direction. If we emit a photon from one box to the other, the photon reaches the second box at time $\Delta t = z/c$. However, the boxes will have picked up an additional speed $\Delta v = a\Delta t = az/c$. So the photon reaching the box will have been redshifted by

$$\frac{\Delta\lambda}{\lambda} = \frac{az}{c^2}$$

We can expect then, that in a uniform gravitational field with same magnitude, we should observe the same redshift, which experiments have verified. A photon travelling down a gravitational field is redshifted after reflection.

1.3 Manifolds

Intuitively, a manifold is a space that locally looks Euclidean, ie looks like $\mathbb{R}^n$. By "looking Euclidean," we don't necessarily need the metric to be Euclidean, rather functions and coordinates work like they should normally. In this case, we call the dimension of the manifold n.

$\mathbb{R}^n$ is obviously a manifold because it looks like $\mathbb{R}^n$ not only locally, but globally. A sphere is of dimension 2 because locally, it looks like $\mathbb{R}^2$. Meanwhile, a line conected perpendicular to a plane is not a manifold, because the dimension is not well defined. A cone is a manifold, however there is a singularity at the tip. Manifolds can be differentiable, but the cone has a problem at the tip, so its not a differentiable manifold. There sphere is more well behaved, so we say it's differentiable.

While the description of a manifold does seem intuitive, we need some heavy lifting to be able to define it more rigorously.

- Let $\phi\colon \mathbb{R}^m \to \mathbb{R}^n$ be a map of sets. We say ϕ is C^p if the pth derivative exists and is continuous. If a map is C^∞ then it is **smooth**.

- We say sets M and N are **diffeomorphic** if there exists a C^∞ map $\phi\colon M \to N$ with inverse $\phi^{-1}\colon N \to M$ that is also C^∞.

- An **open ball** is the set of all points x in $\mathbb{R}^n$ such that $|x-y| < r$ for some $y \in \mathbb{R}^n$ and $r \in \mathbb{R}$, where $|x-y|$ is defined by the euclidean distance $[\sum_i (x^i - y^i)^2]^{1/2}$. Any open set in $\mathbb{R}^n$ can be constructed by the union of open balls.

- A **chart** is a subset U of some set M equipped with a one to one map $\phi\colon U \to \mathbb{R}^n$ such that the image of $\phi(U)$ is open in $\mathbb{R}^n$. We can then say U is an open set in M. A chart is like a coordinate system, we're able to describe any point in $\mathbb{R}_n$ using elements in some abstract set.

- An **atlas** is a collection of charts $\{(U_\alpha, \phi_\alpha)\}$ that satisfies two conditions. First, the union of U_α is M. Second, if any two charts overlap, $U_\alpha \cap U_\beta \neq 0$, then the map $\phi_\alpha \phi_\beta^{-1}$ maps $\phi_{beta}(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$ onto an open set $\phi(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$, where all of these maps are C^∞. This second condition is a way to say the charts are smoothly "sewn together." If there exists two different coordinate systems for the same points, then transitioning between them is smooth and well defined. This allows us to describe smoothness for functions, take derivatives etc, and their behavior shouldn't depend on the coordinate system.

- A **manifold** is a set M with a maximal atlas; meaning it contains every possible chart. This requirement is important, as two different atlas corresopnding to the same space shouldn't be different manifolds.

Example. Consider S^1. The conventional coordinate system using angles $\theta\colon S^1 \to \mathbb{R}$ runs into a problem. If we include either $\theta = 0$ or $\theta = 2\pi$ then we have a closed interval. But if we exclude both, we haven't covered the whole circle. So we need at least two charts to cover the circle.

1.4 Tensor Analysis

Now we'll describe tensors on manifolds. Unlike in flat space, vectors can't be freely moved around, so it makes more sense to describe them as associated with a single point. On a manifold, the set of all vectors associated with a single point is the tangent space to that point.

- The tangent space to a manifold is the space of directional derivative operators along curves that pass through that point. Namely, if $f\colon M \to \mathbb{R}$ then its directional derivative is an element of the tangent space. This way, we can define tangent space without the notion of tangent vectors, which would be circular.

- A good basis to choose for the tangent space is the set of all partials $\{\partial_\mu\}$. This is because

$$\frac{d}{d\lambda} = \frac{dx^\mu}{d\lambda}\partial_\mu$$

so the directional derivative can be expressed as a linear combination of the partials. We call $\frac{dx^\mu}{d\lambda}$ the components of the tangent vector. This basis $\hat{e}_\mu = \partial_\mu$ is the **coordinate basis**.

- Using this definition, we can see that basis vectors transform covariantly

$$\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}}\partial_\mu$$

and similarly, components transform contravariantly

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu}V^\mu$$

This is more general than our definition before, because we drop the need for any Lorentz transform. Here we can define tensors under any change of basis, not just linear transformations.

- On a manifold, all vectors are tangent vectors, which means they are associated with a directional derivative. A vector field is then defines a map from smooth functions to smooth functions over the manifold, by defining a directional derivative at each point.

- Given vector fields X and Y, we define the commutator $[X, Y]$ by how it acts on a function $f(x^\mu)$

$$[X, Y]f = X(Yf) - Y(Xf)$$

- The gradient of a function is a one-form. The set of all gradients of the coordinate functions x^μ form a basis for the cotangent space, so the basis for one-forms is $\{dx^\mu\}$. They transform contravariantly

$$dx^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu}dx^\mu$$

and components of one forms transform covariantly

$$\omega_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}}\omega_\mu$$

Finally, we can generalize our transformation laws by defining tensors. A tensor is a multilinear map of dual vectors and vectors to $\mathbb{R}$. For example, $T(a\omega, cV) = acT(\omega, V)$. A tensor that takes in k vectors and l dual vectors is a type (k, l) tensor. It's easy to see that scalars are $(0, 0)$ tensors and vectors and dual vectors are $(1, 0)$ and $(0, 1)$ tensors accordingly. Some important facts about tensors:

- Suppose T is a (k, l) tensor, then under any general coordinate transformation T behaves as follows:

$$T^{\mu'_1\cdots\mu'_k}{}_{\nu'_1\cdots\nu'_l} = \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}}\cdots\frac{\partial x^{\mu'_k}}{\partial x^{\mu_k}}\frac{\partial x^{\nu_1}}{\partial x^{\nu'_1}}\cdots\frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}}T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l}$$

We say that T transforms k times contravariantly and l times covariantly

- Suppose T is a (k,l) tensor and S is a (m,n) tensor, Then we define the tensor product

$$T \otimes S(\omega^1, \dots \omega^{k+m}, V^1, \dots, V^{l+n}) =$$

$$T(\omega^1, \dots, \omega^k, V^1, \dots, V^l) \times S(\omega^{k+1}, \dots, w^{k+m}, V^{l+1}, \dots V^{l+n})$$

  In general, the tensor product does not commute.

- In index notation, we can write T as Then in index notation, we can write it as $T^{\mu_1 \cdots \mu_k}{}_{\nu_1 \dots \nu^l}$. The basis for the vector space T resides in can be expressed with basis $\hat{e}_{\mu_1} \otimes \dots \otimes \hat{e}_{\mu_k} \otimes \hat{\theta}^{\nu_1} \otimes \dots \otimes \hat{\theta}^{\nu_l}$. The components can be found by acting the tensor T on the basis vectors and dual vectors.

Example. The metric tensor $\eta_{\mu\nu}$ is a $(0,2)$ tensor that takes in two vectors and outputs a scalar. The metric acting on two vectors returns the inner product

$$\eta(V,W) = \eta_{\mu\nu} V^\mu W^\nu = V \cdot W$$

Properties of the inner product should be familiar. Two vectors are orthogonal if their inner product is 0. Note that because the inner product is a scalar, it's invariant under Lorentz transformations. This means that the cartesian basis is always orthogonal under Lorentz transformation.

Additionally, the norm, or "length" of a vector is defined by its inner product with itself $\eta_{\mu\nu} V^\mu V^\nu$. Unlike in Euclidean space, there is no restriction for the norm to be strictly positive. A positive norm is **spacelike**, a negative norm is **timelike** and 0 is **lightlike** or null.

The metric $g_{\mu\nu}$ is an incredibly important tensor. It is symmetric, has nonzero determinant and is used to raise and lower indices. Some of the important roles of the metric include:

- defines path length and proper time
- allows us to define a "shortest distance" between two points
- replaces the Newtonian gravitational field
- allows us to define local inertial frames

We often like to put the metric into a **canonical form** which means it's a diagonal matrix with entries $-1, -1..., -1, 1..., 1, 0...0$. At any point p on a manifold, there exists a coordinate system $x^{\hat{\mu}}$ such that $g_{\hat{\mu}\hat{\nu}}$ is in canonical form and $\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} = 0$. This is called a **local inertial frame**. This is equivalent to saying in curved spacetime, we can always find a local inertial frame with the Minkowski metric.

Example. The Kroneker delta $\delta^\mu{}_\nu$ is a $(1,1)$ tensor. It inputs a vector and a dual vector and outputs a scalar, but can also be thought of as a map from dual vectors to dual vectors and vectors to vectors. In this framing, the Kroneker delta is just the identity map. We can define the inverse metric $\eta^{\mu\nu}$as satisfying the property

$$\eta^{\mu\nu}\eta_{\nu\rho} = \eta^{\nu\mu}_{\rho\nu} = \delta^\mu{}_\rho$$

Some further properties of tensors:

- We can contract a vector by summing over a repeated upper and lower index. With this, we can use the metric to raise and lower indices: $\eta_{\alpha\beta}T^\alpha = T_\beta$ and $\eta^{\alpha\beta}T_\alpha = T^\beta$. This is how you would convert a vector to its dual form and vice versa. In euclidean space, the metric is its own inverse so the gradient is actually both a vector and 1-form at the same time.

- A **symmetric** tensor is one where any change of indices leaves the tensor unchanged. For example, if $S_{\mu\nu\rho} = S_{\nu\mu\rho}$ then we say S is symmetric in the first two indices. A tensor is **antisymmetric** if it changes sign when indices are changed. $A_{\mu\nu\rho} = -A_{\nu\mu\rho}$ is antisymmetric in its first two indices.

- Symmetric and antisymmetric tensors are very useful in that the contraction of a symmetric and antisymmetric tensor along the corresponding indices is always 0. The symmetric part is $T_{(\mu\nu)} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$ and the antisymmetric part is $T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$. We sometimes want to exclude indices from being symmetrized or antisymmetrized so we use the vertical bars. For example, to write the antisymmetrized $R_{\alpha\beta\gamma\delta}$ in only the first and third indices we use $R_{[\alpha|\beta|\gamma]\delta} = \frac{1}{2}(R_{\alpha\beta\gamma\delta} - R_{\gamma\beta\alpha\delta})$

- For a $(1,1)$ tensor, we write the trace by leaving off the indices (just the contraction along the repeated index)

$$X = X^\mu{}_\mu$$

  We can similarly define the trace of a $(0,2)$ or $(2,0)$ tensor by raising/lowering into a $(1,1)$ tensor and then contracting. Note that in this case, the trace is not just the naive sum over the diagonal; the components change when raising/lowering the index. The reason behind this is because the sum over the diagonal is not a true scalar; ie it's not invariant under coordinate transform.

2 Riemmannian Geometry

2.1 Covariant Derivative

The partial derivative doesn't behave like a tensor under coordinate transformation, so we need to define a new object, the **covariant derivative**, that serves as a partial derivative on manifolds and transforms like a tensor. Additionally, it should reduce to the partial derivative in inertial frames. There are a few motivational details to keep in mind:

- The partial derivative ∂_μ maps (k,l) tensor fields to $(k,l+1)$ tensor fields. The covariant derivative ∇ should do the same.

- Any good derivative should obey both linearity and the Liebniz rule:

$$\nabla(T+S) = \nabla T + \nabla S$$

$$\nabla(T \otimes S) = (\nabla T) \otimes S + T \otimes (\nabla S)$$

- To obey the Liebniz rule, ∇ can be written as a partial derivative plus some linear transformation. This gives us the definition of the covariant derivative:

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda$$

where $\Gamma^{\nu}_{\mu\lambda}$ are known as the **connection coefficients** or **Christoffel symbols**. They are there to make sure the covariant derivative transforms like a tensor, and they themselves are not tensors because they don't obey the transformation laws.

- The covariant derivative commutes with contraction, so $\nabla_\mu(T^{\lambda}{}_{\lambda\rho}) = (\nabla T)_\mu{}^{\lambda}{}_{\lambda\rho}$. Additionally, it should reduce to the partial derivative for scalars $\nabla_\mu\phi = \partial_\mu\phi$
- One can show using the last property that the covariant derivative of one forms is given by
$$\nabla_\mu\omega_\nu = \partial_\mu\omega_\nu - \Gamma^{\lambda}_{\mu\nu}\omega_\lambda$$

Of course, the connection is not unique. Any tensor satisfying the Liebniz rule, linearity, and reducing to the partial derivative can be a covariant derivative, each of which can have a connection associated with it. In general relativity, however, there are several additional constraints:

- The connection is **torsion free**, meaning $\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$. This is equivalent to saying the covariant derivative of a scalar field is independent of the order of differentiation, so $\nabla_\mu\nabla_\nu\phi = \nabla_\nu\nabla_\mu\phi$.
- The connection is **metric compatible**, or $\nabla_\rho g_{\mu\nu} = 0$. This further means that the covariant derivative commutes with raising and lowering of indices $g_{\mu\lambda}\nabla_\rho V^{\lambda} = \nabla_\rho(g_{\mu\lambda}V^{\lambda}) = \nabla_\rho V_\mu$
- There exists a unique connection satisfying this; the Levi-civita connection
$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

2.2 Parallel Transport

In Euclidean space, we often say a vector doesn't have an associated "position" because we can freely move it around and it will remain unchanged. However, this is not true in curved space. The idea of moving a vector along a path while keeping it constant is called **parallel transport**. Notably, parallel transport depends on the path taken between points.

In fact, there is no "natural" way to compare vectors in different tangent spaces, since there are different paths to take. Thus, we can only compare two vectors if they are in the same tangent space. For example, the relative velocities of two particles on different points of a manifold is not well defined!

- By "keeping the vector constant," we mean its components must be constant. For an arbitrary tensor, this condition is $\frac{d}{d\lambda}T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = \frac{dx^\mu}{d\lambda}\frac{\partial}{\partial x^\mu}T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0$. Then to make this a proper tensor equation, it makes sense to replace the partial derivative with the covariant derivative so we define the **directional covariant derivative** to be
$$\frac{D}{d\lambda} = \frac{dx^\mu}{d\lambda}\nabla_\mu$$
The equation for parallel transport is then
$$\left(\frac{D}{d\lambda}T\right)^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} \equiv \frac{dx^\sigma}{d\lambda}\nabla_\sigma T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0$$
More commonly, we will be parallel transporting vectors, so the equation is of form
$$\frac{d}{d\lambda}V^\mu + \Gamma^{\mu}_{\sigma\rho}\frac{dx^\sigma}{d\lambda}V^\rho = 0$$

- We define a geodesic as the curve along which a tangent vector is parallel transported. This gives us $\frac{D}{d\lambda}\frac{dx^\mu}{d\lambda} = 0$ and expanding this gives us

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0$$

  The so-called **geodesic equation**. We can see that in euclidean space, the equation reduces trivially with the connection coefficients $\Gamma^\mu_{\rho\sigma}$ vanishing so we have $\frac{d^2x^\mu}{d\lambda^2} = 0$ gives you the equation of a straight line.

- This definition of a geodesic is equivalent to the path that maximizes the proper time, which one can show using variational calculus. This makes sense since in spacetime, such paths are "locally straight".

- For timelike paths with four velocity $U^\mu = \frac{dx^\mu}{d\tau}$ the geodesic equation can equivalently be written as

$$U^\lambda \nabla_\lambda U^\mu = 0$$

2.3 Curvature

We already have some notions of curvature by the differences between Euclidean and more general spaces. For example, whether or not parallel transport leaves vectors unchanged on a closed loop. The **Riemann Tensor** will formalize these ideas into a single object.

- Consider an infinitesimal loop defined by two vectors A^μ and B^ν. Imagine parallel transporting some vector V^μ along A^μ, then along B^ν, then back along A^μ and back along B^ν. In curved space the vector will return to its original position and have been changed, and likewise parallel transporting it along the opposite path will also result in changing the vector. The Riemann tensor tells us the difference between parallel transporting the vector along the first path and the second path.

- Note that parallel transport is independent of coordinates, so it must be a linear transformation of a vector involving one upper and one lower index. It must also depend on the vectors we used to define the loop, so there must be two additional lower indices to contract with them.

- Finally the tensor must be antisymmetric in the two indices.

- As per our definition above, the Riemann tensor should be related to commutator of the covariant derivative

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu} - T^\lambda \mu\nu \nabla_\lambda V^\rho$$

  where $R^\rho{}_{\sigma\mu\nu}$ is the Riemann tensor and $T^\lambda_{\mu\nu}$ is the Torsion tensor.

- Expanding, and eliminating terms that cancel when using the antisymmetric property, we have

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$$

  This holds for any connection, whether or not it is metric compatible or torsion free.

- Using our intuition above, we can think of the Riemann tensor as a map from 3 vector fields to another vector field. The first two arguments should be the fields specifying the "direction" along our infinitessimal loop, and the result should act on the third field (the one we're parallel transporting). Using the definition from the commutator, we have

$$\boxed{R(X,Y)Z = \nabla_X \nabla_y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z}$$

  The notation ∇_X is the covariant derivative along a vector field X so we can write it in component form as $\nabla_X = X^\mu \nabla_\mu$. The last term with the commutator vanishes when X and Y are coordinate basis vector fields becanse $[\partial_\mu, \partial_\nu] = 0$.

Now we will analyze a few properties of the Riemann Tensor in local inertial coordinates; ie the Christoffel symbols drop. It's easier to work with it in all lower indices $R_{\rho\sigma\mu\nu} = g_{\rho\lambda} R^\lambda{}_{\sigma\mu\nu}$:

- The Riemann tensor is antisymmetric in the first two indces $R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}$ and its last two indices $R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}$ It is also invariant under changing the first and second pairs of indices: $R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}$.

- The sum of cyclic permutations of the last three indices vanishes: $R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0$. This is equivalent to saying the antisymmetric part of the last 3 indices is 0: $R_{\rho[\sigma\mu\nu]} = 0$ and equivalently $R_{[\rho\sigma\mu\nu]=0}$.

- We can imagine the tensor as a symmetric matrix $R_{[\rho\sigma][\mu\nu]}$, where each pair corresponds to an antisymmetric matrix. Then we have $\frac{1}{2}[\frac{1}{2}n(n-1)][\frac{1}{2}n(n-1)+1]$ independent components where we used the number of components of an $m \times m$ symmetric matrix and $n \times n$ antisymmetric matrix. Finally, the totally antisymmetric component vanishes so this reduces the number of independent components to $\frac{1}{12}n^2(n^2-1)$

- Taking the covariant derivative, we see the sum of cyclic permutations also vanish so $\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0$. By antisymmetry in the first two indices, we can write

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0$$

  known as the **Bianchi Identity**.

- We define the **Ricci Tensor** as the contraction

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$$

  For the Christoffel connection, the Ricci tensor is symmetric $R_{\mu\nu} = R_{\nu\mu}$.

- The contraction of the Ricci Tensor is the **Ricci Scalar**:

$$R = R^\mu{}_\mu = g^{\mu\nu} R_{\mu\nu}$$

Finally we define the **Einstein Tensor** as $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$.

2.4 Killing Vectors

Symmetries allow us to solve problems that would otherwise have been painfully difficult. Even if a situation is slightly deviated from a symmetry, we can still account for this through perturbations. We say a manifold M possesses a symmetry if the metric is perserved under a transformation mapping M onto itself. A symmetry of the metric is called an **isometry**.

- Lorentz transformations and translations are isometries of Minkowski space. This is because the metric coefficients $\eta_{\mu\nu}$ are independent of coordinates x^μ. For some coordinate σ:

$$\partial_\sigma g_{\mu\nu} = 0 \implies x^\sigma \to x^\sigma + a^\sigma \text{is a symmetry}$$

- For a timelike path, the geodesic equation can be written as $p^\lambda \nabla_\lambda p^\mu = 0$. By metric compatibility, we can lower the index and expand the covariant derivative to get

$$p^\lambda \partial_\lambda p_\mu - \Gamma^\sigma_{\lambda\mu} p^\lambda p_\sigma = 0$$

The first term determines how the components of momentum change along the path $p^\lambda \partial_\lambda p_\mu = m\frac{dp_\mu}{d\tau}$ while the second term reduces to $\frac{1}{2}(\partial_\mu g_{\nu\lambda})p^\lambda p^\nu$ so the geodesic equation can be written as

$$m\frac{dp_\mu}{d\tau} = \frac{1}{2}(\partial_\mu g_{\nu\lambda})p^\lambda p^\nu$$

But if the metric is independent of the coordinates x^σ, then the isometry implies $\frac{dp_\sigma}{d\tau} = 0$ so the momentum four vector is conserved along a geodesic.

It is important to note that while invariance of the metric implies a symmetry, the converse is not. There are four types of translations and six types of Lorentz transformations, which is ten total symmetries. But this is greater than the number of dimensions the metric can be independent of. Instead, we'll need a more formal procedure:

- Suppose $g_{\mu\nu}$ is independent of x^σ. Then let us define $K = \partial_\sigma$, which in component notation is

$$K^\mu = (\partial_\sigma)^\mu = \delta^\mu_\sigma$$

Then K^μ generates the isometry, namely any transformation of this form can be expressed as motion in the direction of K^μ.

- Any four momentum along the direction of K^ν then obeys $p_\sigma = K^\nu p_\mu = K_\nu p^\nu$. Careful! We've specified σ as a specific direction in which the metric is invariant of, so p_σ is in fact a scalar; it's the component of the four momentum in the direction of K^ν

- The scalar being constant along the path is equivalent to saying the directional derivative along the geodesic is 0, so $p^\mu \nabla_\mu (K_\nu p^\nu) = 0$. This is because of the chain rule, because covariant derivatives of scalars reduces to the partial derivative. By expanding this equation, we get:

$$\nabla_{(\mu} K_{\nu)} = 0 \implies p^\mu \nabla_\mu (K_\nu p^\nu) = 0$$

where the left side is **Killing's equation** and vector fields K that satisfy it are called **Killing vector fields**.

- Killing vector fields on a manifold are in one-to-one correspondence with the continuous symmetries of the metric on that manifold.

3 Gravitation

3.1 Minimal Coupling

The motivation behind the use of manifolds was because of several of the postulates established before. The Einstein equivalence principle stated that in small regions of spacetime, the laws of physics reduce that to special relativity, and that local experiments cannot detect the existence of gravity. This is very similar to our ability of finding locally inertial coordinates on manifolds; ie where the metric reduces to the Minkowski metric and first derivatives vanish. The **Minimal Coupling Principle** allows us to use this fact to generalize laws of physics in curved spacetime.

1. Take a law of physics in inertial coordinates in flat spacetime
2. Convert it into a coordinate-invariant tensor equation
3. Assert the law holds in curved spacetime

Example. Consider the equation of motion of a free particle travelling in flat space along a parametrized straight line $x^\mu(\lambda)$. This is expressed with the equation

$$\frac{d^2x^\mu}{d\lambda^2} = 0$$

We can rewrite this using the chain rule: $\frac{d^2x^\nu}{d\lambda^2} = \frac{dx^\mu}{d\lambda}\partial_\nu\frac{dx^\mu}{d\lambda}$, but the partial derivative is not a tensor, so we replace it with the coordinate invariant form; the covariant derivative:

$$\frac{dx^\nu}{d\lambda}\nabla_\nu\frac{dx^\mu}{d\lambda} = \frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda}$$

which is just the geodesic equation.

Example. We should also check to see if curvature alone is sufficient to explain gravity in the Newtonian limit. We assume particles are moving slowly, and that the gravitational field is weak and static.

By moving slowly, we demand the velocity to be much less than one (working in natural units), $\frac{dx^i}{d\tau} \ll \frac{dt}{d\tau}$, which simplifies the geodesic equation to be

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dt}{d\tau}\right)^2 = 0$$

In a static field, the components of $g_{\mu\nu}$ are unchanging with time so $\partial_0 g_{\mu\nu} = 0$, which means the christoffel symbols simplify to

$$\Gamma^\mu_{00} = -\frac{1}{2}g^{\mu\nu}\partial_\lambda g_{00}$$

Finally, in a weak field, we can represent the metric as a small perturbation in the minkowski metric:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

where $|h_{\mu\nu}| \ll 1$. We can then write the christoffel symbols to first order in $h_{\mu\nu}$, giving us $\frac{d^2x^\mu}{d\tau^2} = \frac{1}{2}\eta^{\mu\nu}\partial_\lambda h_{00}\left(\frac{dt}{d\tau}\right)^2$ But since $\partial_0 g_{00} = 0$, the timelike component of the geodesic equation is just $\frac{d^2t}{d\tau}^2 = 0$. Meanwhile, the spacial component gives us

$$\frac{d^2x^i}{dt^2} = \frac{1}{2}\partial_i h_{00}$$

This reduces to the Newtonian limit if we let $h_{00} = -2\mathbf{\Phi}$. You can see this matches Poisson's equation $\nabla^2\mathbf{\Phi} = 4\pi G\rho$ exactly.

3.2 The Stress-Energy Tensor

We'll now introduce the **stress-energy tensor $\mathbf{T}^{\mu\nu}$**, which is the tensorial equialent to mass.

- We can think of the components of $T^{\mu\nu}$ as the flux of four momentum p^μ across a surface of constant x^ν. The temporal component is T^{00}, the flux of p^0 energy across the x^0 time dimension. This is just the rest frame energy density.
- T^{i0} are the momentum flux across the temporal dimension, which is just the rest momentum density. Meanwhile, T^{0i} is the energy flux across a spacial dimension. To show these are equivalent, recall that energy and momentum in relativity are $E = \gamma mc^2$ and $p = \gamma mv$. Then the energy density ϵ and momentum density g are related by $g = \frac{1}{c^2}\epsilon v$. If we consider energy E crossing a cuboid of thickenss L across area A in time Δt, then the energy flux is $\frac{E}{A\Delta t} = \frac{E}{AL/v} = \epsilon v$, which is just the momentum density up to a constant factor. Thus, in the rest frame, the T^{i0} and T^{0i} components are equivalent.
- The T^{ij} components reduce to the classical stress tensor $t^{ij}$from newtonian mechanics. Non diagonal terms represents shear: the ith component of force per unit area (pressure) exerted across a surface normal to the jth direction. For a fluid, this could represent forces such as viscocity. The diagonal terms give us the pressure $p_i = T^{ii}$.
- For a perfect fluid at rest, the force exerted across a surface is always along the normal and the same in all directions, equivalent to pressure $\mathbf{t}^{ij} = p\delta^{ij}$.

Example. Dust is a collection of particles at rest with respect toeach other, travelling at constant four velocity U^μ, We can choose to move to inertial frame where $U^\mu = (1,0,0,0)$. This is called the **comoving frame**.

- The **number current** is $N^\mu = nU^\mu = n(u^0, \mathbf{u}^i)$ where n is the number density (particles per unit volume). The components of the number current is the number of particles travelling through unit area per unit time which is the particle flux.
- The rest **mass density** is $\mu_0 = m_0 n$ where m_0 is the rest mass of a single particle. If there is internal energy, then we have $\mu = \mu_0(1+\delta)$ where δ is the internal energy fraction. The **momentum density** is $\rho^i = m_0 n\mathbf{u}^i$.
- In the rest frame, ρ is the $\mu = 0$, $\nu = 0$ component of the tensor product $p^\mu \otimes N^\nu$ so we conjecture that the stress energy tensor for dust is

$$T^{\mu\nu}_{textdust} = p^\mu N^\nu = mnU^\mu U^\nu = \rho U^\mu U^\nu$$

Example. A perfect fluid has isotropic rest frame pressure p and energy density ρ. Because of isotropy, the stress energy tensor is diagonal

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}$$

We can further simplify this expression depending on the type of fluid we are dealing with by using the equations of state $p = p(\rho)$

- For dust, we saw that $p = 0$
- An isotropic gas has $p = \frac{1}{3}\rho$
- The vacuum energy has $p = -\rho$ which means the stress tensor is proportional to the metric $T^{\mu\nu} = -\rho_{\text{vac}}\eta^{\mu\nu}$ Energy in a vacuum is irrelevent in special relativity because energy can be always be changed up to a scalar value. However, this fact is not true in general relativity because all energy couples to gravity.

A covariant expression that reduces to this in the rest frame is then

$$T^{\mu\nu} = (\rho + p)U^{\mu}U^{\nu} + p\eta^{\mu\nu}$$

Some additional properties of the stress tensor:

- As we've seen, $T^{\mu\nu}$ is symmetric. Additionally, the stress tensor is conserved, meaning it is "divergence free."
$$\partial_{\mu}T^{\mu\nu} = 0$$
- Expanding this equation and contracting in the U_{ν} component, we obtain $U_{\nu}\partial_{\mu}T^{\mu\nu} = -\partial_{\mu}(\rho U^{\mu}) - \partial_{\mu}U^{\mu} = 0$. In the nonrelativistic case,the four velocity is $U^{\mu} = (1, v^i)$, where $|v^i| \ll 1$ and $p \ll \rho$. The second term drops and we are left with
$$\partial_t\rho + \nabla \cdot (\rho\mathbf{v}) = 0$$
which is also known as the continuity equation
- Meanwhile, using the projection operator $P^{\sigma}{}_{\nu} = \delta^{\sigma}_{\nu} + U^{\sigma}U_{\nu}$ and applying it to conservation of the stress tensor, eliminating all terms first order in $\mathbf{v}$ which are small, we have
$$\rho\left[\partial_t\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v}\right] = -\nabla p$$
which is the Euler equation from fluid mechanics.

3.3 The Einstein Equations

Derivation of the Einstein Equations

Poisson's equation $\nabla^2\Phi = 4\pi G\rho$ relates gravitational fields to the mass density. However, we've been building up to the idea that gravity can be explained by curvature alone. Thus, a more fundamental equation relating curvature and mass.

First, we should replace gravitational potential with a tensorial object; the metric immediately comes to mind, as in the newtonian limit we were able to reproduce gravity by perturbations in the metric. Meanwhile, the left side is second order in the derivatives of the potential, so we should look for something similarly second order in the metric that describes curvature; the Einstein tensor.

The einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}$ always obeys

$$\nabla^\mu G_{\mu\nu} = 0$$

so we may postulate an equation in the form

$$G_{\mu\nu} = \kappa T_{\mu\nu}$$

since it would automatically satisfy conservation of stress energy. Contracting both sides gives us $R = -\kappa T$, so we can rewrite the Ricci tensor as

$$R_{\mu\nu} = \kappa(T_{\mu\nu} - \frac{1}{2}T g_{\mu\nu})$$

To determine the constant factor, we'll take the case of a dust in the Newtonian limit. Working in the rest frame, the energy momentum tensor is

$$T_{\mu\nu} = \rho U_\mu U_\nu$$

where $U^{mu} = (U^0, 0, 0, 0)$. But in the rest frame, we can take $U_0 \approx 1$, so we have $T_{00} = \rho$. Note that in the rest frame, the T_{00} term dominates all the other terms, so the trace to lowest order is just $T = g^{00}T_{00} = -T_{00} = -\rho$ which gives us $R_{00} = \frac{1}{2}\kappa\rho$. To solve for the metric, we need to evaluate $R_{00} = R^\lambda{}_{0\lambda 0}$ but $R^0{}_{000} = 0$ so we only need the spacial components. Now we have

$$R^i_{0j0} = \partial_j\Gamma^i_{00} - \partial_0\Gamma^i_{j0} + \Gamma^i_{j\lambda}\Gamma^\lambda_{00} - \Gamma^i_{0\lambda}\Gamma^\lambda_{j0}$$

The second term vanishes for static fields, while the third and fourth terms are second order in perturbation so they can be neglected. Thus,

$$R_{00} = R^i{}_{0i0} = \partial_i\left[\frac{1}{2}g^{i\lambda}(\partial_0 g_{\lambda 0} + \partial_0 g_{0\lambda} - \partial_\lambda g_{00})\right]$$

But to first order and for static fields, the temporal derivatives vanish, reducing the equation to

$$= -\frac{1}{2}\delta^{ij}\partial_i\partial_j h_{00} = -\frac{1}{2}\nabla^2 h_{00} = \frac{1}{2}\kappa\rho$$

which gives us Poisson's equation if $\kappa = 8\pi G$. Thus, we've determined Einstein's equation as

$$\boxed{R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = 8\pi G T_{\mu\nu}}$$

Using $R = -8\pi GT$, it's sometimes helpful to write it in this form

$$R_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right)$$

where we can see in a perfect vacuum, the equations reduce to $R_{\mu\nu} = 0$

Because the metric is

4 Gravitational Waves

Recall the form of the metric in the linearized theory $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$.

- The einstein equations give us $2(R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R) = 16\pi GT_{\alpha\beta}$
- Introducing the trace-reversed metric perturbation, $\overline{h}_{\alpha\beta} \equiv h_{\alpha\beta} - \frac{1}{2}h$ and expanding the einstein equation, we can simplify this to $\Box\overline{h}_{\alpha\beta} = \overline{h}_{\alpha\beta,\mu}^{,\mu} = -16\pi GT_{\alpha\beta}$. The box is the d'Alembert operator and is the equivalent of the laplacian in spacetime. This nicely gives us a wave equation for $\overline{h}$.
- The solution for this wave equation is $\overline{h}_{\mu\nu} = Re(A_{\mu\nu}e^{ik_\alpha x^\alpha})$ where $A_{\mu\nu}$ gives the amplitude and k_α is the wavevector $(w, \mathbf{k})$. From the d'Alembert operator, we have $k_\alpha k^\alpha = 0$, or k is a null vector. This means gravitational waves propogate at the speed of light.
- We can choose a Lorentz Gauge to simplify this further (what does this mean?) with $\overline{h_{\mu\nu}^{;\nu}} = 0$. This gives us $A_{\mu\alpha}k^\alpha = 0$ meaning these are transverse waves (why?)
- Starting with the Symmetric 4x4 tensor, we have 10 independent coordinates. Being tranverse and traceless takes away 4 degrees of freedom (why?) and the gauge condition takes away 4 more so we are left with 2 polarizations. Imposing all conditions gives us the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & a & b & 0 \\ 0 & b & -a & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- Both linear and circular polarization like light (This section needs work)

Now we'll derive the metric from a weakly gravitating source. Suppose we have a static point mass M located at point $r = 0$ at all times in the weak field limit. In the body's frame we have $|h_{\mu\nu}| \ll 1$.

- Because the body is stationary, the stress energy tensor has components $T_{00} = \delta(x)M$, and all other components are 0.
- From the einstein equation and poisson equations, we have $\overline{h}_{00} = \frac{4M}{r}$ and all other components are 0.
- Converting from $\overline{h}$ back to h and substituting into the equation for the metric $ds^2 = (\eta_{\mu\nu} + h_{\mu\nu})dx^\mu dx^\nu$ we obtain the metric:

$$ds^2 = 1\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 + \frac{2M}{r}\right)(dx^2 + dy^2 + dz^2)$$

5 Black Holes

Rev. John Mitchell found in 1783 that an object with $500_{\odot}$ but of same density will have its escape velocity to be greater than the speed of light. Then "all light emitted from such a body would be made to return towards it."

- Every mass has a radius such that if the mass is small enough than this radius, it will collapse into a black hole. This is the **Schwarzchild Radius**. $R_s = \frac{2GM}{c^2}$

5.1 Schwarzchild Metric

The Schwarzchild Metric has two forms:

- The timelike form $d\tau^2 = (1 - \frac{2M}{r})dt^2 - \frac{dr^2}{(1-\frac{2M}{r})} - r^2(d\theta^2 + \sin^2\theta d\phi^2)$
- The spacelike form $d\tau^2 = (1 - \frac{2M}{r})dt^2 - \frac{dr^2}{(1-\frac{2M}{r})} - r^2 d\Omega^2$
- We can see the metric reduces to flat spacetime in the case where $r \to \infty$ and $m \to \infty$
- Upon examining the temporal component of the metric, we that its less than that of the temporal component in flat spacetime, so this properly predicts gravitational redshift. Similarly, the spacial component is larger than that in flat spacetime, so there's spacetime curvature.

5.2 Coordinates Systems

These are three commonly used frames:

- Free-Float Frame: Frame free-floating in space, think unpowered spacecraft. Locally its flat minkowski space. For observers on shells, they see spacecraft infalling towards the black hole.
- Spherical-Shell Frame: Frame standing still on a shell of fixed radius (eg: Earth's frame). For small time lapses and regions, special relativity applies. Notice that inside a black hole, dr becomes time like while dt becomes spacelike. This means staying on a shell at constant radius is impossible, so this frame cannot be used to describe events inside a black hole.
- Schwarzchild Frame: Global book-keeping coordinates. Idea is that neither spherical shell frame nor free float frame can know all information so both send information to a Global frame that puts it together.

A common result from black holes is that from the perspective of someone outside of the black hole, we see an object travel into the black hole past the event horizon. But from the perspective of the falling object, it takes infinite time to reach the event horizon!

- We can calculate the Riemman tensor in the shell frame: $R_{trtr} = -\frac{2M}{r^3} = -R_{\theta\phi\theta\phi}$ and $R_{t\theta t\theta} = R_{t\phi t\phi} = \frac{M}{r^3} = -R_{r\theta r\theta} = -R_{r\phi r\phi}$
- Meanwhile in the free falling frame we have $R_{trtr} = -\frac{2M}{r^3}(1 - \frac{2M}{r})^{-1}$. This isn't singular at $r = 2M$ but it is at $r = 0$. The curvature invariant (also known as the Kretschmann scalar) $R = R_{trtr}R^{trtr} = \frac{48M^2}{r^6}$. This gives a measure of the strength of the gravitational field using curvature and this also blows up to infinity at $r = 0$.

5.3 Orbits

Because the Schwarzchild metric is time independent and spherically symmetric, we must have killing vectors $\xi_t = (1, 0, 0, 0)$ and $\xi_\phi = (0, 0, 0, 1)$. For any four-velocity u, $\xi \cdot u$ is constant. combining this with $p^\mu = mu^\mu$, we obtain $\frac{E}{m_0 c^2}$ is constant along a path.

- Radial Infall: Let $d\tau$ be the proper time for an observer at distance r from the center of a black hole (shell frame). At $r \to \infty$, we have $\frac{E_\infty}{m_0 c^2} = \frac{dt}{d\tau}$. But E_∞ is a constant. So any trajectory starting at rest from $r = \infty$ will have same energy per unit mass throughout. This gives us $d\tau^2 = (1 - \frac{2M}{r})^2 dt^2$.

- With this expression we can obtain the velocity recorded by a Schwarzchild observer: $\frac{dr}{dt} = -(1 - \frac{2M}{r})(\frac{2M}{r})^{\frac{1}{2}}$. At $r = 2M$, this goes to 0 so we see that the bookeeper observes the particle slowing down as it approaches the event horizon, taking infinite time to reach there.

- If we instead moved to the frame of the shell observer, we can obtain from the Schwarzchild metric: $\frac{dr_{shell}}{dt_{shell}} = -(\frac{2M}{r})^{\frac{1}{2}}$. For an observer at the event horizon, they would see the object falling at the speed of light! The energy observed also depends on the distance from the center (while E_∞ is constant, E_{shell} is not).

- Meanwhile, for the free falling frame, we can actually compute the time it takes for the observer to reach $r = 0$. Using the chain rule, $\frac{dr}{d\tau} = \frac{dr}{dt}\frac{dt}{d\tau}$ we can obtain $\tau = \frac{4}{3}M$. In seconds this is $\tau_{second} = \frac{4}{3}\frac{M}{M_{odot}}(5\mu sec)$. Interestingly, the time scales linearly with mass, so for larger black holes it actually takes more time to fall! For reference, Sagittarius A, the supermassive black hole in the center of the galaxy, it takes roughly 1 minute to fall. Meanwhile, TON618, one of the largest black holes takes 11 days to fall to the center.

Now we'll go over orbits. In the case with $r > 20R_s$, the newtonian approximation is still accurate. For $r \leq 6M$, no free orbits are possible. The mass spirals in but still may exert enough energy to escape. Beyond $r \leq 2M$, no escape is possible. Inside the horizon, as mentioned before, space and time components are interchanged. Travelling to smaller r is as "inevitable" as outside a black hole we travel to greater t.

- Recall the conserved quantity from earlier $\xi_t \cdot u^t$. We call this the "energy per unit mass" and we'll just set this equal to $\frac{E}{m}$. The free fall case assumes this is one but this is not true generally. Write the angular momentum per unit mass as $\frac{L}{m} = r^2 \frac{d\phi}{d\tau}$. Using these, we have the equation of motion

$$\left(\frac{dr}{d\tau}\right)^2 = \left(\frac{E}{m}\right)^2 - \left(1 - \frac{2M}{r}\right)\left[1 + \frac{(L/M)^2}{r^2}\right] \equiv \left(\frac{E}{m}\right)^2 - \left(\frac{V}{m}\right)^2$$

  . This gives us an effective potential of

$$\left(\frac{V(r)}{m}\right)^2 \equiv \left(1 - \frac{2M}{r}\right)\left[1 + \frac{(L/m)^2}{r^2}\right]$$

  In the Newtonian case, we have an elliptical orbit. But for black holes, orbits tend to spend more time at smaller radii, the additional time near the inner edge of the orbit is the "**dwell time**."

- Stable circular orbits at V_{min} and unstable orbits occur at V_{max}. Beyond V_{max} the orbit precesses because of the dwell time. (this section needs work)

5.4 Massless Particle

We'll now cover orbits of massless particles. This is especially important for describing the behavior of light near a black hole. The procedure for massive particles won't work here for several reasons.

- The spacetime interval for massless particles is 0 (why?). Solving for the coordinate velocity of light in these coordinates gives us $r\frac{d\theta}{dt} = \pm(1-\frac{2M}{r})^{\frac{1}{2}}$ tangentially and $\frac{dr}{dt} = \pm(1-\frac{2M}{r})$ radially. This looks like it means the speed of light can even approach 0 but this actually isn't a problem because this interpretation is unphysical. Even if the Schwarzchild observer "sees" the speed of light as this, any shell observer still sees it at constant c.

- At $r = \infty$, the special relativity case we have $p = p_\infty$. The angular momentum L can be written using this momentum as $L = bp_\infty$ where b is known as the **impact parameter**. (Why?) Using the expression for P in special relativity, we can write $b = \frac{L/m}{E/m}$

- The impact parameter conveniently allows us to evaluate the speed even with 0 mass:
  The radial velocity: $(\frac{dr}{dt})^2 = (1-\frac{2M}{r})^2 - (1-\frac{2M}{r})^3\frac{b^2_{photon}}{r^2}$
  The tangential velocity : $r\frac{d\phi}{dt} = \frac{b_{photon}}{r}(1-\frac{2M}{r})$
  Thus the total speed is $v_{photon} = (1-\frac{2M}{r})[1+\frac{2Mb^2}{r^3}]^{\frac{1}{2}}$.
  With a similar technique, we can verify that in the shell coordinates, the speed of the particle is constant 1.

- The equivalent equation of motion in the shell frame is $\frac{1}{b^2}(\frac{dr_{shell}}{dt_{shell}})^2 = \frac{1}{b^2} - \frac{(1-\frac{2M}{r})}{r^2}$. $\frac{1}{b^2}$ plays the role of the energy squared while the second term is the equivalent of the effective potential.

- This potential has no stable orbits. A photon may enter a circular "knife edge" orbit before either escaping or plunging.

6 Cosmology

6.1 The Present Universe

In the last 100 years, there were many important observations about our current universe.

- Hubble discovered the universe is expanding in 1929. The speed at which galaxies recessed from us is proportional to their distance $v = Hr$ where H is the hubble constant.

- In 1965, Penzias and Wilson discovered the cosmic microwave background has temperature $T = 273°K$. It is also highly isotropic and homogeneous.

- Matter density upper bounds at $10^{-29}g/cm^3$ and lower bound at $5\times10^{-31}g/cm^3$. The Baryon number Photon number ratio is $10^-8 - 10^{-10}$

- From stars, galaxies and globular clusters, the Helium-4 mass fraction is 0.2-0.3, while Deutereum is 3×10^{-5}

- There is a high degree of local structure, with galaxies, galaxy clusters and galaxy filaments. Supermassive objects like galaxy filaments, voids and walls put the cosmological principle into question.

- Dark energy (or vacuum energy) is at 0.69, corresponding to 69% of total energy density in the universe.

Now we'll derive the Einstein Equations for a Robertson-Walker Universe, and show that this satisfies the isotropy and homogeneity conditions at all points.

- Consider an infinite, homogeneous gas cloud with mass density ρ. Choose an arbitrary point A as the origin and consider a thin spherical shell centered around A with radius r. By conservation of energy, we have $(\frac{\dot{r}}{r})^2 + \kappa(r) = \frac{8G\rho}{3}$. κ is $\frac{k}{r^2}$ where k is the total energy.
- By using the First Law of Thermodynamics and assuming constant entropy, we can obtain $\frac{d\rho}{dt} + \frac{3}{r}\frac{dr}{dt}(\rho + p) = 0$. Taking the first derivative of the previous line and combining, we have $\frac{\ddot{r}}{r} + \frac{4\pi G}{3}(\rho + 3p) = 0$.
- Choosing a coordinate comoving with the gas cloud, we can define $\mathbf{r(t)} = a(t)\mathbf{r_0}$. $a(t)$ is the scale factor that depends only on time (no dependence on space because isotropy). From this, we can rewrite our earlier equations to obtain
  $\frac{\ddot{a}}{a} + \frac{4\pi G}{3}(\rho + 3p) = 0$
  $(\frac{\dot{a}}{a})^2 + \frac{\kappa(r)}{a^2} = \frac{8G\rho}{3}$. These are the the einstein equations in component form.
- The continuity equation of the stress energy tensor can similarly be rewritten $\dot{\rho}+3(\frac{\dot{a}}{a})(\rho+p) = 0$.
- These equations can be further simplified using equations of state.
  Dust: $p = 0$
  Radiation: $p = \frac{1}{3}\rho$
  Vacuum Energy: $p = -\rho$

6.2 History of the Universe

Hubble's discovery of the expansion of the universe and Penzias & Wilson's discovery of the cosmic microwave background suggest the existence of the universe in a "hot, dense" state in the very beginning, before expansion. This is the popularly named "**Big Bang**" model (Gamow 1948, Dicke, Peebles 1965).

Furthermore the existence of General Relativity combined with the energy dominance condition on matter (energy is positive, and energy-momentum doesn't travel faster than light) implies the existence of a cosmological singularity (Penrose, Hawking, Geroch 1966). (Why?)

- Starting with the Friedmann universe, we can do a convenient change of variables by replacing the cumbersome $(1 - kr^2)^{-1}$ with χ. This is known as the comoving distance. As the universe expands, the proper distance between locations change with time but χ remains constant $D(t) = a(t)\chi$.

$$ds^2 = -dt^2 + a^2(t)[d\chi^2 + f^2(\chi)(d\theta^2 + \sin^2\theta d\phi^2)]$$

  where $f(\chi) = r$. This gives us nice forms for χ:

  i $k = 1$ Closed universe: $f(\chi) = \sin(\chi)$
  ii $k = 0$ Flat universe: $f(\chi) = \sinh \chi$

iii $k = -1$ Open universe: $f(\chi) = \chi$

In the case where $f(\chi) = \sin(\chi)$ and $0 \leq \chi \leq 2\pi$ we have a closed universe. For $f(\chi) = \chi$ and $0 \leq \chi \leq \infty$ this is a flat universe. Finally, for $f(\chi) = sinh(\chi)$ then this is an open universe. (What do these mean?)

- Substituting the equatins of state into the continuity equation we can obtain scaling factors of the energy density with a In the flat universe,

  i dust : $\rho_m \sim a^{-3}$
  ii radiation : $\rho_\gamma \sim a^{-4}$
  iii vacuum : $\rho \sim const$

- Solving the einstein equation with the Friedmann metric we obtain the following:

$$G_{00} \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} - \frac{\Lambda}{3} = \frac{8\pi G\rho}{3}$$

  This is the energy conservation equation, which relates the rate to energy density and

$$G_{ii} \to 3\left(\frac{\ddot{a}}{a}\right) = -4\pi G(\rho + 3p)$$

  This is the acceleration equation.

- If we solve these equations using the equations of state in the flat case ($k = 0$) we obtain the following:

  i In a matter dominated universe: $a \sim t^{\frac{2}{3}}$
  ii Radiation dominated: $a \sim t^{\frac{1}{2}}$
  iii Vacuum dominated: $a \sim e^{Ht}$

6.3 Singularity

A big bang corresponds to a scale factor of $a = 0$. We can ask the question if our universe was ever at this point ie: does it take finite time to reach $a = 0$ in the past?

- First, assume the universe is radiation-dominated (which it would've been at that time) and with $\Lambda = 0$ Then from the G_0^0 equation, we obtain $a^2 = \left(\frac{32}{3}\pi GB\right)^{\frac{1}{2}}(t - t_0)$ where $\rho = Ba^{-4}$. This means it takes finite time to reach $a = 0$

- A similar analysis for the G_i^i equation gives us $\frac{\dot{a}}{a} \to \infty$ at $t = t_0$, which implies this time has infinite expansion rate.

- Causality laws: the speed of sound is always less than the speed of light imply that $\rho \to \infty$ at this time.

Since the universe began at $R = 0$ a finite time in the past, this makes the Big Bang inevitable. This point in time is called the cosmological singularity. Penrose and Hawking's singularity theorems (1973) show that the universe must have been a singularity even if the conditions of isotropy and homogeneity were relaxed.

6.4 The FLRW Universe

- We define the proper distance between 2 galaxies as the integral

$$d(t) = a(t) \int_0^r \frac{dr}{\sqrt{1-kr^2}} = a(t) \begin{cases} \sin^{-1} r & \text{for } k=1 \\ r & k=0 \\ \sinh^{-1} r & k=-1 \end{cases}$$

  Meanwhile the proper velocity is

$$v(t) \equiv \dot{d}(t) = \frac{\dot{a}(t)}{a(t)} d(t) = H(t)d(t)$$

  this is Hubble's law, where $H_0(t_0)$ is the present value of the hubble constant.

- Let r_{ph} be the coordinate of the most distant object observable, with the earliest light coming from time t_{min}. For light rays, we can set $ds = 0$ giving us

$$\chi_p = c \int_{t_{min}}^{t_0} \frac{dt}{a(t)} = \int_0^{r_{ph}} \frac{dr}{\sqrt(1-kr^2)}$$

  The closed universe model has $r_{ph} = \sin(\chi_p)$ which is a finite value, so there exists a horizon beyond which you cannot see past (why?). This is called the particle horizon.

6.5 The CMB

6.6 The Early Universe